

# Wood Carbon Based Single-Atom Catalyst for Rechargeable Zn-Air Batteries

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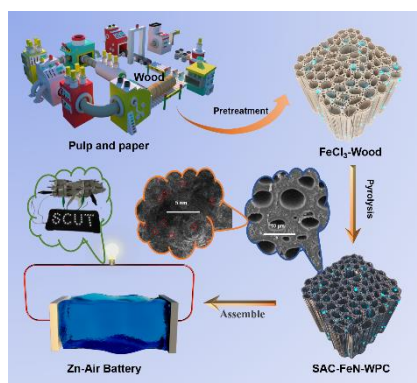
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**Abstract:** Low-cost and efficient oxygen reduction reaction (ORR)/oxygen evolution reaction (OER) bifunctional electrocatalysts are vital for the applications of rechargeable Zn-air batteries (ZABs). Given the high catalytic activity of single-atom catalysts (SACs), preparing SACs on a large-scale for ZABs is desirable, but remains challenging. Herein, in-situ formation of single-atom Fe-N-C catalysts on plate wood-based porous carbon is achieved via facile Lewis acid pre-treatment and carbonization process. Lewis acid  $\text{FeCl}_3$  pretreatment on the cell wall of wood not only produces abundant microchannels, but also successfully introduces atomically dispersed Fe-N active species into the hierarchical structure. Such uniformly dispersive SACs on the hierarchical structure enhance ORR/OER performance and durability. ZAB using the catalyst in the cathode shows a high power density ( $70.2 \text{ mW cm}^{-2}$ , at quasi-solid-state) and long-term stability. This work provides a new path for the large-scale preparation of high-performance SACs.

## TOC GRAPHICS

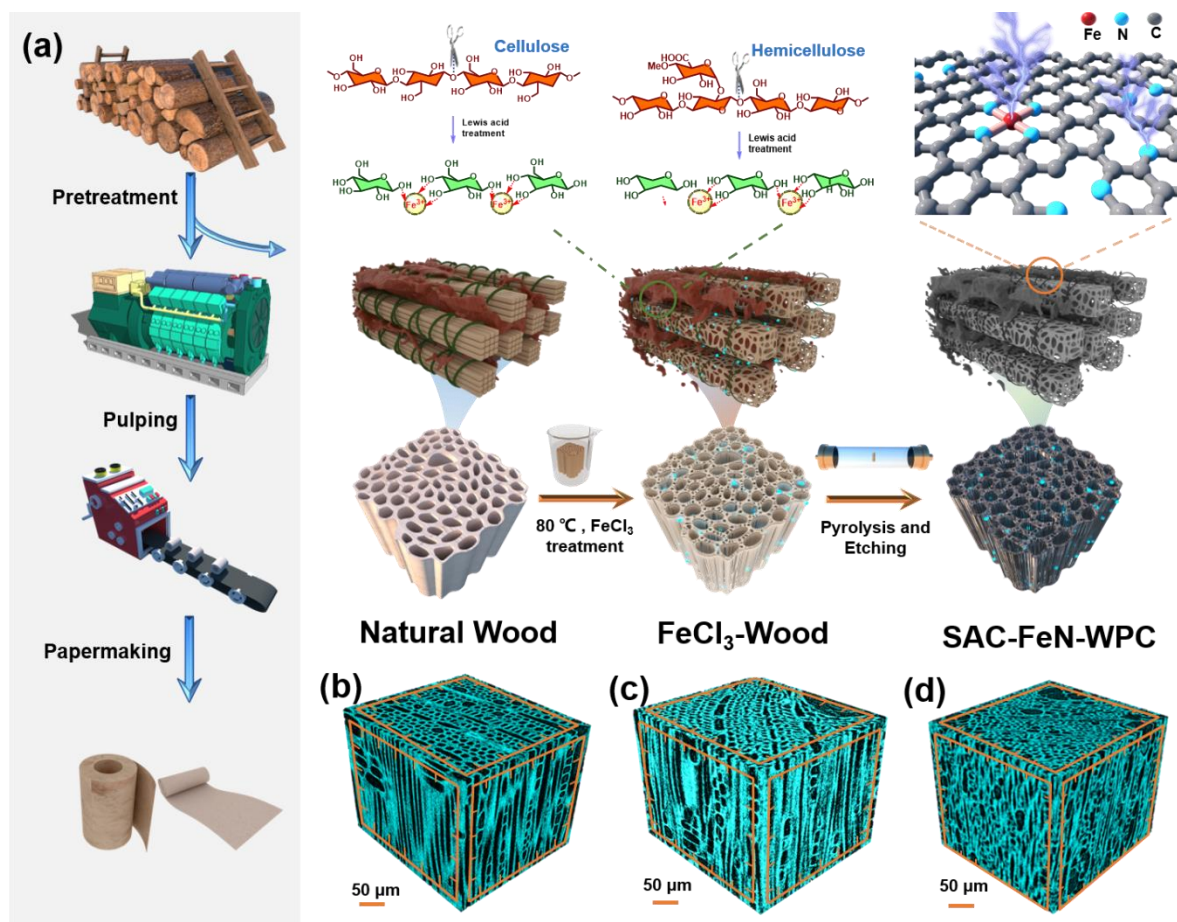


The bottleneck of the application of ZABs is the sluggish kinetics and high overpotentials of the oxygen reduction reaction (ORR) and oxygen evolution reaction (OER).<sup>1-4</sup> Though noble metal catalysts such as Pt/C, RuO<sub>2</sub>, and IrO<sub>2</sub> can effectively improve the reaction kinetics, the high cost and unsatisfactory stability fuel the exploration of efficient and durable non-precious metal catalysts. Among them, metal-nitrogen-carbon (Fe-N-C) catalysts have been considered as a promising candidate to substitute noble metal-based materials.<sup>5,6</sup> Yet, their catalytic performance is still far from satisfactory. Increasing the intrinsic reactivity and the number of active sites is the consensus towards highly effective catalysts. The emergence of single-atom catalysts (SACs) opens a new path for developing effective single atom Fe-N-C catalysts for ORR/OER given their unique physicochemical properties.<sup>7-9</sup> SACs consist of isolated metal atoms dispersed or anchored on matrix materials.<sup>10-12</sup> Uniform distribution of SACs, strong bonding between SACs and the substrate,<sup>13-25</sup> and effective mass transfer/electron transport are the prerequisites for achieving high catalytic performance.<sup>26</sup> However, the poor control over dispersion and uniformity of atomic Fe and dopant heteroatoms entail the dependency on the precision equipment and complex processes, which confines the large-scale application of single atom Fe-N-C catalysts in ZABs.

Developing substrate materials with hierarchical and porous structures can boost their functionalities via improving the accessibility of active sites and mass transport.<sup>20,23,24,27</sup> Wood naturally has a hierarchical structure, providing a desirable interconnected and ordered channels for mass transport.<sup>28-30</sup> The exploitation of such natural resources as the carrier for SACs is desirable for the practical application. The anchoring of metals to the support, metal-support interactions (geometric effects, charge transfer), confinement, and surface functional groups can generate single atoms that can exist on carbon-supported catalysts.<sup>31,32</sup> Wood-based catalysts anchored with metal particles or doped with heteroatoms have reported with high electrocatalytic performances.<sup>30, 33-35</sup> However, utilizing wood as the carriers for SACs

and their application in ZABs remains unexplored. To achieve high electrocatalytic performances, uniform dispersion of single atoms in the wood microchannels and the high availability of the single atoms are required.

Herein, we propose a novel and effective strategy for the large-scale preparation of plate-shaped single-atom catalyst in a two-in-one manner (**Figure 1a**). Lewis acids have been widely used in the early stage of pulping industry pre-treatment to hydrolyse wood into 5-hydroxymethylfurfural and other products.<sup>36-39</sup> In this work,  $\text{FeCl}_3$  is adapted to simultaneously partially hydrolyse the cellulose and hemicelluloses in the wood, which not only leaves abundant nanopores but also embeds large number of  $\text{Fe}^{3+}$  ions in the microchannels. The coordinated  $\text{Fe}^{3+}$  was further reduced and stabilized on the N-doped carbonized wood-based carbon after carbonization. The as-prepared single-atom Fe-N-C catalyst loaded wood-based porous carbon (SAC-FeN-WPC) was directly used as a self-supporting electrode material without the use of a binder and carbon cloth for ZABs. Owing to the hierarchical porous structure and uniformly dispersive single atom on the wood-based carbon matrix, SAC-FeN-WPC delivered an enhanced ORR/OER reactivity with an ORR half-wave potential of 0.85 V (vs. reversible hydrogen electrode, RHE), and an overpotential  $\Delta E$  ( $E_{\text{OER}} - E_{\text{ORR}}$ ) of 0.78 V. ZAB constructed with the SAC-FeN-WPC as cathode electrode showed a high power density ( $70.2 \text{ mW cm}^{-2}$ , at quasi-solid-state) and long-term stability. Such an approach successfully introduced single atom Fe-N-C catalysts onto the resourceful wood, proving a promising strategy for the cost-effective and large-scale preparation of SACs.



**Figure 1.** Fabrication and microstructure of SAC-FeN-WPC. (a) Schematic illustration of the fabrication procedure of SAC-FeN-WPC. Microscopic 3D structure of (b) Natural wood, (c) FeCl<sub>3</sub>-wood, and (d) SAC-FeN-WPC.

Figure 1a illustrates the preparation procedure of SAC-FeN-WPC. First, Lewis acid treatment was conducted to introduce iron into the wood (Figure S1a) by impregnating the natural wood into FeCl<sub>3</sub> solution (Figure S1). The resultants were denoted as FeCl<sub>3</sub>-wood. After this process, the smooth surface of natural wood became much rougher, as indicated by the scanning electron microscopy (SEM) images of natural wood and FeCl<sub>3</sub>-wood (Figure S2 and S3). The FeCl<sub>3</sub>-treated wood was then pyrolyzed under an Ar-NH<sub>3</sub> atmosphere, followed by HCl etching to remove inactive iron species. Surprisingly, after the pyrolysis and etching process, a plate phase SAC-FeN-WPC with a thickness of about 0.1 cm was obtained, along with good robustness (Figure S1b and c). This process produces bulk wood-based carbon-

supported SACs, showing great potential in the large-scale preparation of effective SACs for rechargeable ZABs.

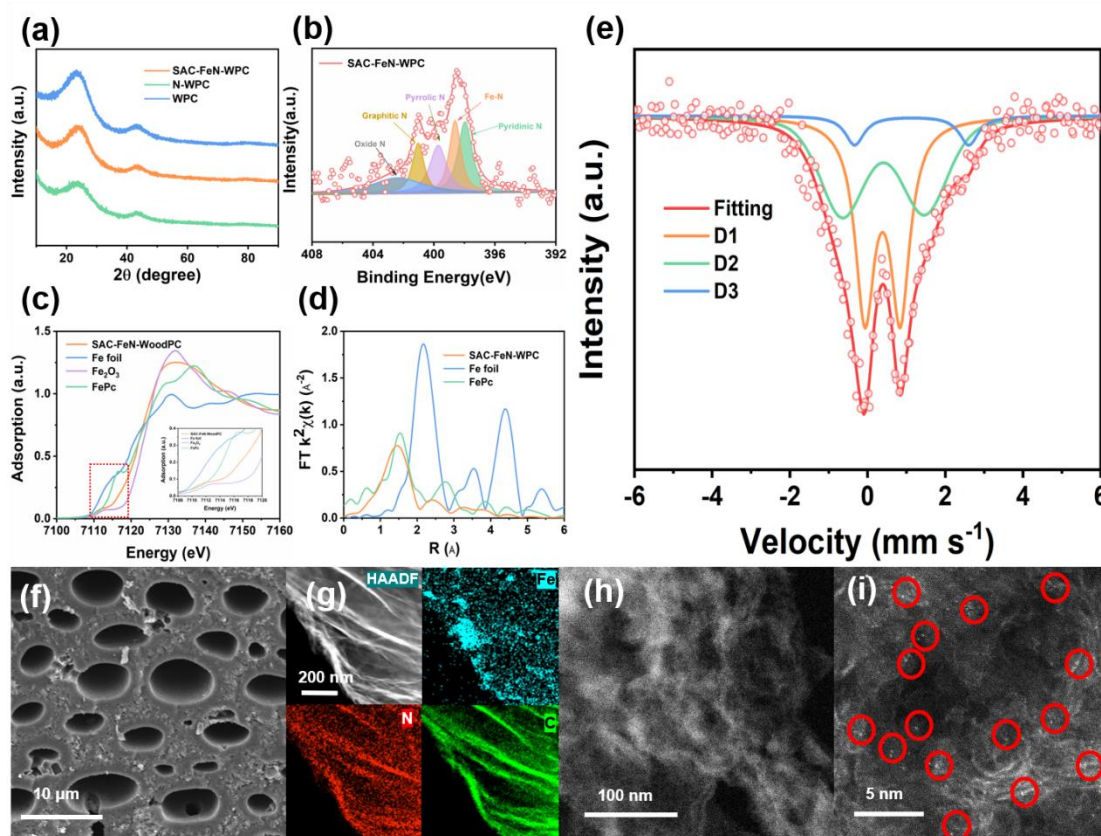
The evolution of pores was evaluated by Nano-CT scanners and SEM. Figure 1b-d show the nano-CT images of natural wood, FeCl<sub>3</sub>-wood, and SAC-FeN-WPC, respectively. It is found that the hierarchical structure of wood remains intact after the FeCl<sub>3</sub>-treatment and more intriguingly, the SAC-FeN-WPC exhibits an enhanced porous structure even after the pyrolysis. There is no difference in the surface morphology and porous structure between untreated wood carbonized in the Ar-NH<sub>3</sub> environment (N-WPC) and wood (Figure S4). However, the SEM images of SAC-FeN-WPC disclose a much rougher surface with a large amount of shallow nanoholes on the pore walls (Figure S5), indicating the unique role of FeCl<sub>3</sub> pre-treatment in the formation of nanopores in the wood. The underlying pore-forming mechanism lies in the break-down of the  $\beta$ -1, 4-glycosidic bonds in cellulose and hemicelluloses macromolecules during Lewis acid hydrolysis happens on the surface of raw wood in an acidic aqueous solution, leading to a decrease in the degree of polymerization and consequently the appearance of pores in the wood (Figure 1c).<sup>37-39</sup>

To reveal the unique role of FeCl<sub>3</sub> plays in pore-forming, we chose two other Lewis acids, namely ZnCl<sub>2</sub> and AlCl<sub>3</sub>, and conducted the acid pre-treatment under the same condition to prepare ZnCl<sub>2</sub>-wood and AlCl<sub>3</sub>-wood, following by the identical pyrolysis and etching process to obtain ZnCl<sub>2</sub>-N-WPC and AlCl<sub>3</sub>-N-WPC. Normally, the acid pre-treatment of wood with different Lewis acids results in a varying extent of hydrolysis of cellulose and hemicelluloses given the difference in acid strength. Partial hydrolysis leads to a loose structure, which holds promise in generating micro/nano-pores on the cell wall during the carbonization process. As the strongest Lewis acid selected, pre-treatment using ZnCl<sub>2</sub> gives rise to the roughest surface but nearly collapsed skeleton as indicated by the nano-CT and SEM images (Figure S6a-c). In comparison, the natural hierarchical structure of WPC remains in AlCl<sub>3</sub>-wood, and only several randomly dispersed shallow holes are observed on

its surface (Figure S6d-f). Nonetheless, in FeCl<sub>3</sub>-wood, not only the layered structure but also a rough surface with uniformly distributed shallow nanopores are identified (Figure S3).

Brunauer-Emmett-Teller (BET) measurement was conducted to further evaluate the surface area and pore-size distribution of as-prepared samples. After the carbonization, the surface area of N-WPC increases (Table S1), along with a combination of type I and type IV with a hysteresis loop at  $P/P_0 \approx 0.95$ , indicating the presence of micropores and mesopores (Figure S7a). Woods treated with Lewis acid, namely ZnCl<sub>2</sub>, AlCl<sub>3</sub>, and FeCl<sub>3</sub>, possess much larger surface areas than N-WPC and drastically different pore structures. In the case of the increase in the surface area, as discussed above, the hydrolysis of cellulose and hemicelluloses will shear the compacted interaction between cellulose/hemicelluloses microfibrils and lignin, resulting in a rough surface and loosely structured wood-based substrate. Hence, the BET surface area of the Lewis acid-treated wood significantly increases after the carbonization process (Table S1). The order of surface area further affirms the proportional relationship between the strength of Lewis acid strength and pore-making ability, which is attributed to the roughness of wood treated by different Lewis acids (Figure S3 and S6). The excessive hydrolysis of wood by ZnCl<sub>2</sub> leads to the largest BET surface area of ZnCl<sub>2</sub>-N-WPC. Unfortunately, ZnCl<sub>2</sub> fails to retain the layered structure of wood. In contrast, FeCl<sub>3</sub> preserves the natural hierarchical structure to the greatest extent under the premise of ensuring the formation of pores, laying a foundation for the preparation of plate-like SAC-FeN-WPC. The porosity of acid-treated wood samples is superior to that of non-treated ones (Figure S7c-d and Table S1). All acid-treated woods possess the co-existence of micropores and mesopores.<sup>30</sup> In detail, AlCl<sub>3</sub>-N-WPC shows fewer micropores, while the prominent richness in both micropores and mesopores in the plate SAC-FeN-WPC verifies its hierarchical porous structure and the unique pore-forming function of FeCl<sub>3</sub> treatment.





**Figure 2.** Characterizations of SAC-FeN-WPC. **(a)** XRD patterns for SAC-FeN-WPC, N-WPC, and WPC. **(b)** N 1s high-resolution XPS spectrum of SAC-FeN-WPC. **(c)** Fe K-edge XANES spectra of SAC-FeN-WPC and reference samples. **(d)** Fourier transformed (FT)  $k^3$ -weighted  $\chi(k)$ -function of the EXAFS spectra for Fe K-edge. **(e)** Room-temperature  $^{57}\text{Fe}$  Mössbauer spectrum of the  $^{57}\text{Fe}$ -enriched SAC-FeN-WPC. **(f)** SEM image of SAC-FeN-WPC. **(g)** HAADF-STEM images of SAC-FeN-WPC and the corresponding EDS mapping images for iron, nitrogen, carbon. **(h, i)** Aberration-corrected HAADF-STEM images of the SAC-FeN-WPC with the bright points representing iron atoms.

In addition to its vital role in constructing porous structure,  $\text{FeCl}_3$  is a Fe source for the in-situ formation of Fe single atom. The X-ray diffraction (XRD) patterns of wood-based porous carbon (WPC), N-WPC, and SAC-FeN-WPC show two broad diffraction peaks at around  $26^\circ$  and  $44^\circ$ , corresponding to the (002) and (101) crystal planes of graphitic carbon, respectively (**Figure 2a**).<sup>21,23</sup> No peaks belonging to iron species are detected in the XRD patterns due to

the trace iron content (~0.8 wt%, determined by the inductively coupled plasma (ICP) measurement). The Raman spectrum of SAC-FeN-WPC (Figure S8) exhibits a strong D band, indicating a highly disordered carbon structure with a large number of defects. The bonding states of Fe, N, and C in N-WPC and SAC-FeN-WPC were investigated by X-ray photoelectron spectroscopy (XPS). As shown in Figure 2b, the high-resolution N 1s XPS profile of SAC-FeN-WPC shows five peaks at 402.4, 401.0, 399.7, 398.6 and 398.0 eV, corresponding to oxide N, graphitic N, pyrrolic N, Fe@N bonding (N atoms coordinating with Fe atoms), and pyridinic N, respectively.<sup>20</sup> Graphitic N contributes to an enhanced ORR catalytic performance via improving the limiting current density of the catalyst.<sup>40</sup> Higher contents of nitrogen species are found in SAC-FeN-WPC (Figure S9 and Table S1), which is attributed to the chelation between Fe<sup>3+</sup> and the oxygen-containing functional groups in hydrolysates from cellulose/hemicelluloses, namely glucose and xylose. The chelating Fe<sup>3+</sup> can provide more in-situ coordination between atomic Fe and NH<sub>3</sub> during pyrolysis process, since N atoms are easier to bond with metal atoms rather than C atoms.<sup>41-43</sup> Beyond that, the porous structure is also more favorable for N doping, which is conducive for the formation of single atomic Fe on nitrogen-doped wood-based carbon.

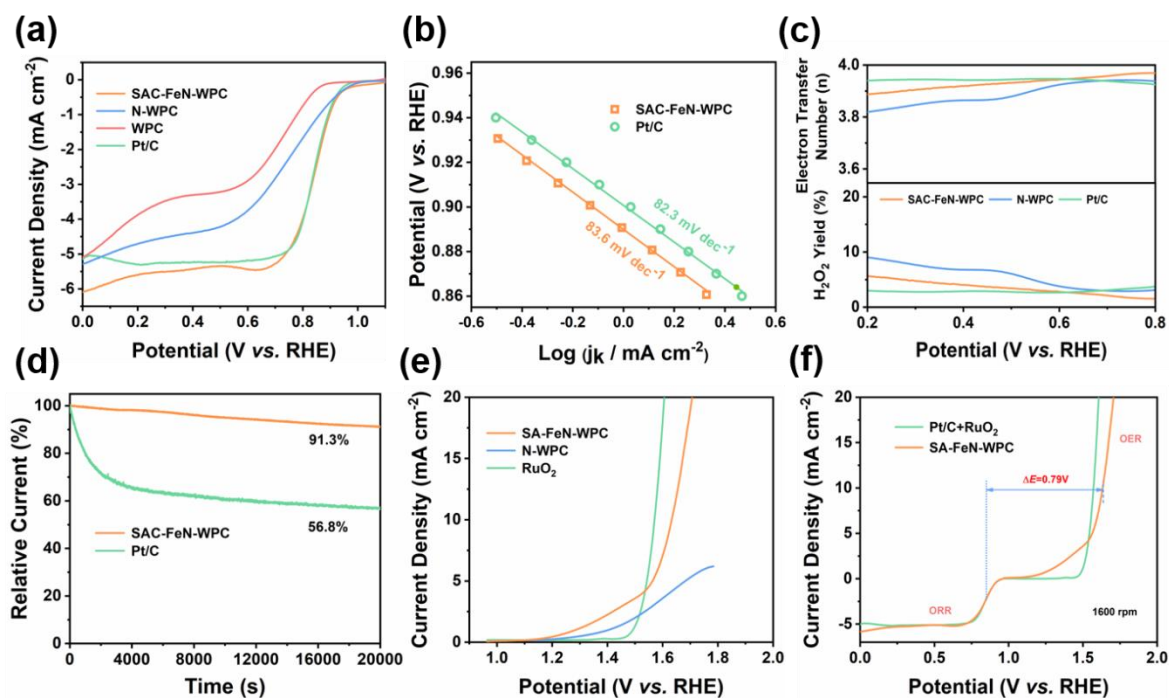
In the SEM and high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of SAC-FeN-WPC (Figure 2f and g), there are no nanoparticles or nanoclusters over the wood-based carbon matrix, while the energy-dispersive X-ray spectroscopy (EDS) clearly demonstrates the uniformly distributed Fe, N, and C elements over the entire SAC-FeN-WPC. The presence of homogeneously dispersed N and Fe on the carbon skeleton is further confirmed by SEM elemental mapping results (Figure S10). The Aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (AC-HAADF-STEM) was applied to detect the atomic Fe within the plate SAC-FeN-WPC catalysts (Figure 2h and i).<sup>4</sup> The white dots are also spotted in other regions of SAC-FeN-WPC (Figure S11), suggesting the uniform distribution of single Fe atoms.

Element-selective X-ray absorption fine structure (XAFS) measurements at the Fe K-edge were performed to further investigate the chemical state and coordination environment of Fe species. The results show that the Fe species are present in the form of dispersed single Fe atoms combined with nitrogen to form FeN<sub>4</sub> structure.<sup>15</sup> Figure 2c shows the normalized X-ray absorption near edge structure (XANES) of the Fe K-edge for SAC-FeN-WPC and Fe foil and FePc references. The pre-edge peak and near-edge adsorption energy of SAC-FeN-WPC are intermediate to those of Fe<sub>2</sub>O<sub>3</sub> and FePc, indicating that the oxidation state of Fe after the in-situ reduction of the samples by pyrolysis is +2~+3, which is also confirmed by the high-resolution Fe 2p<sub>3/2</sub> XPS profile of the sample (Figure S12).<sup>44-46</sup> The Fourier transformed (FT) k<sup>3</sup>-weighted  $\chi(k)$  function of the EXAFS of the samples was conducted to acquire more information about their atomic structures. As can be observed in Figure 2d, the FT-EXAFS profile of SAC-FeN-WPC exhibits a single main peak at 1.42 Å, corresponding to the Fe-N scattering paths. Compared to the Fe foil, SAC-FeN-WPC shows no noticeable Fe-Fe peak (2.20 and 4.42 Å), indicating the successful isolation of Fe atoms. Furthermore, the wavelet transforms (WT) of the Fe K-edge extended EXAFS oscillations of SAC-FeN-WPC also confirms the atomic dispersion of Fe species. As shown in Figure S13, the WT contour plot of SAC-FeN-WPC shows only one intensity maximum at approximately 4.00 Å which is very close to that of FePc (4.00 Å), attributing to Fe-N coordination. In addition, SAC-FeN-WPC shows no intensity maximum related to the Fe-Fe (7.5 Å) coordination, unlike the Fe foil, FePc, and Fe<sub>2</sub>O<sub>3</sub> references. The HAADF-STEM results (Figure 2i and S11) show that in SAC-FeN-WPC, all the Fe species are atomically disperse and there is no detectable aggregation. These results demonstrate that the state of Fe in SAC-FeN-WPC is alike to that in FePc. The coordination configuration of Fe atoms in SAC-FeN-WPC was investigated by quantitative EXAFS curve fitting (Figure S14). The best-fitting analysis clearly shows the main peak caused by the coordination of the first shell of Fe-N (Figure S15), which is different from that of the Fe foil and Fe<sub>2</sub>O<sub>3</sub> (Figure S16 and S17).

Mössbauer spectroscopy is a powerful analysis for identifying iron species, especially for ion with similar coordination environment but different electronic states. As shown in Figure 2e, the Mössbauer spectrum of SAC-FeN-WPC can be deconvoluted into three doublets (D1, D2 and D3), while no sextet and singlet components are observed, indicating the absence of zero-valent iron crystalline phases.<sup>20,46-48</sup> In detail, the doublet D1 with small isomer shift (IS,  $\sim 0.38$ ) and quadrupole-splitting (QS,  $\sim 0.91$ ) is typically attributed to square-planar Fe-N<sub>4</sub> coordination with Fe<sup>2+</sup> in a low spin state, in agreement with the XANES and EXAFS analysis (Table S2).<sup>46</sup> Meanwhile, other fractions of middle spin (D2 doublet, Fe<sup>2+</sup> or Fe<sup>3+</sup>,) and high spin (D3 doublet, Fe<sup>2+</sup>) iron species are detected in the SAC-FeN-WPC. The relatively large QS in high spin state may demonstrate a coordination environment of Fe-N<sub>4</sub> analogue with one or two axial ligands to form a deformed octahedral structure.<sup>[47]</sup> These results further assure the underlying pore- and SACs-forming mechanisms in SAC-FeN-WPC. Precisely, the hydrolysis of wood exposes oxygen-containing functional groups on the surface of pore walls, providing chelated sites for Fe<sup>3+</sup> which will coordinate with NH<sub>3</sub> and further be reduced to atomic Fe atom in Ar-NH<sub>3</sub> atmosphere during the carbonization process. The distribution of single Fe-N-C catalysts is uniform, given the even roughness of the surface of the wood after FeCl<sub>3</sub>-treatment (Figure S3). The hierarchical porous carbon-support single atom Fe-N-C catalysts with interconnected micropores and mesopores are favourable to enhance catalytic performance given the abundant active sites and interior channels for the mass transport of oxygen and the electrolyte.<sup>49</sup> Further, the microporous structure between the particles can serve as an electrolyte reservoir.<sup>26</sup>

The electrocatalytic properties of SAC-FeN-WPC towards ORR were evaluated by a typical three-electrode system in O<sub>2</sub>-saturated 0.1 M KOH. The current densities of the samples are normalized to the electrode disk area. The linear sweep voltammetry (LSV) curves were recorded at a rotating rate of 1600 rpm. Remarkably, SAC-FeN-WPC exhibits an outstanding ORR activity with a half-wave potential ( $E_{1/2}$ ) of 0.85 V (vs. RHE), which is

much higher than those of N-WPC, WPC and SAC-FeN-WPC before etching, and also comparable to that of 20 wt% Pt/C catalyst (Figure 3a and S18). The onset potential of SAC-FeN-WPC is 0.99 V (VS. RHE), which is almost the same as that of 20 wt% Pt/C and higher than that of N-WPC (0.93 V vs. RHE). Furthermore, SAC-FeN-WPC presents the largest reduction current density in the diffusion-controlled region, which indicates its enhanced mass- and electron-transfer capability derived from the hierarchically porous structure and uniformly dispersed single-atom Fe-N-C catalysts. The excellent ORR activity of SAC-FeN-WPC is also manifested from its smaller Tafel slope of 83.6 mV dec<sup>-1</sup>, which is comparable to that of 20% Pt/C (82.3 mV dec<sup>-1</sup>) (Figure 3b). The good linearity of the K-L plots of SAC-FeN-WPC suggests a near four-electron transfer for the ORR process (Figure S19), which is further confirmed by the H<sub>2</sub>O<sub>2</sub> yields from the ring currents (Figure 3c). The superior ORR performance of SAC-FeN-WPC can be attributed to the synergetic effect of its highly active FeN sites and hierarchical porous structure.<sup>30,49,50</sup> The durability of the as-prepared catalysts was investigated by chronoamperometry tests (Figure 3d). The current density of SAC-FeN-WPC remains 91.3% after 20000 s. By contrast, the commercial Pt/C catalyst suffers from a rapid decrease (~56.8%) in the current density. The excellent durability of SAC-FeN-WPC makes it promising for application in ZABs. Through the optimization of different treatment temperatures and concentrations, the best process parameters are obtained (Figure S20).



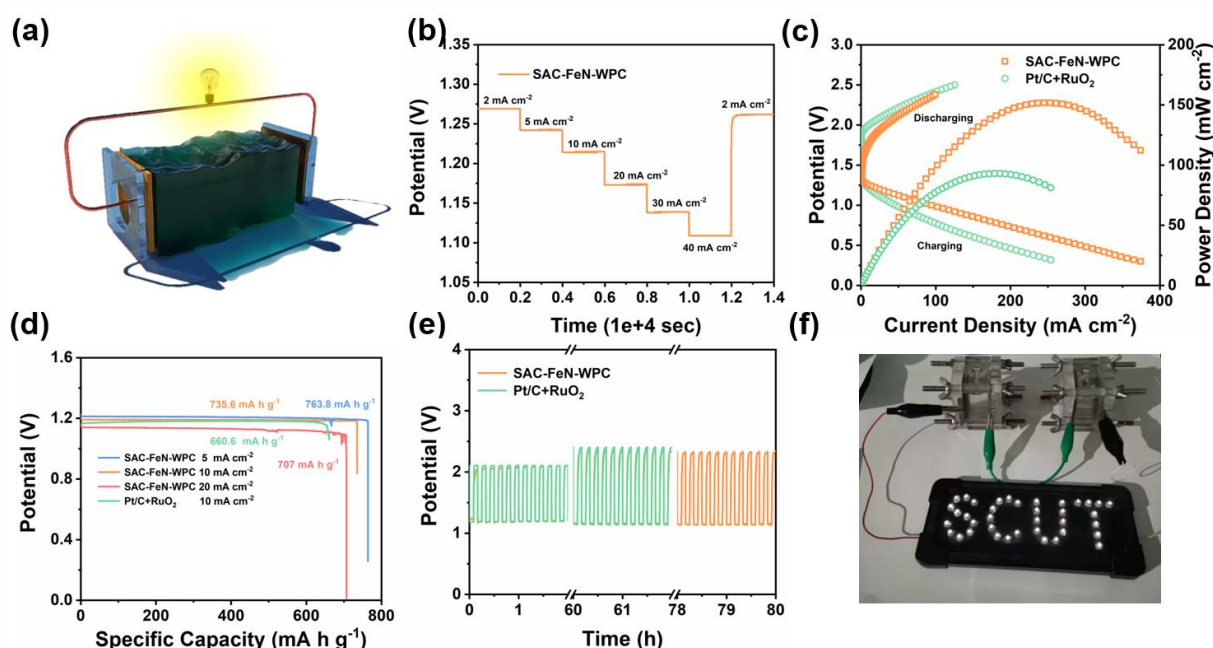
**Figure 3.** Electrocatalytic performances of the catalysts. **(a)** LSV ORR curves for SAC-FeN-WPC, N-WPC, WPC and 20% Pt/C in O<sub>2</sub>-saturated 0.1 M KOH solution at the rotating rate of 1600 rpm. **(b)** Tafel plots of SAC-FeN-WPC and 20% Pt/C in alkaline conditions. **(c)** H<sub>2</sub>O<sub>2</sub> yields and electron reduction pathways for SAC-FeN-WPC and Pt/C. **(d)** Chronoamperometric curves for the ORR of SAC-FeN-WPC and Pt/C at +0.564 V (vs RHE) in an O<sub>2</sub>-saturated 0.1 M KOH aqueous solution at 1600 rpm. **(e)** LSV OER curves of SAC-FeN-WPC, N-WPC and RuO<sub>2</sub> at 1600 rpm in 0.1 M KOH as the electrolyte with IR correction. **(f)** Polarization curves of the as-prepared catalysts for bifunctional catalytic activity.

The OER catalytic performances were assessed in N<sub>2</sub>-saturated 0.1 M KOH. The benchmark RuO<sub>2</sub> catalyst and N-WPC catalyst were tested for comparison. As illustrated by OER LSV curves in Figure 3e, SAC-FeN-WPC exhibits the optimal OER activity with a low overpotential of 400 mV (comparable to that of RuO<sub>2</sub>, 340 mV) at 10 mA cm<sup>-2</sup>. The good OER performance of SAC-FeN-WPC can be attributed to the catalytic effect of pyridine nitrogen.<sup>51-53</sup> We tested the contents of N species before and after OER i-t test and found that

the content of pyridine nitrogen decreased significantly after OER i-t test, while the contents of other nitrogen species were almost unchanged (Figure S21). By comparing the ORR and OER performances of samples treated with different Lewis acids under the same process conditions, the importance of hierarchical porous characteristics and the advantages of single-atom are proved (Figure S22). The overall ORR/OER activities of SAC-FeN-WPC were assessed by calculating the potential difference between  $E_{j=10}$  for OER and  $E_{1/2}$  for ORR ( $\Delta E = E_{j=10} - E_{1/2}$ ). The SAC-FeN-WPC presents a  $\Delta E$  value of 0.79 V, which is close to those of Pt/C for the ORR ( $E_{1/2}$ ) and  $\text{RuO}_2$  for the OER ( $E_{j=10}$ ) ( $\Delta E = 0.72$  V) (Figure 3f). The  $\Delta E$  value of SAC-FeN-WPC is comparable to those of other iron or cobalt-based SACs reported recently (Table S3).<sup>54-69</sup>

To evaluate the practical application of SAC-FeN-WPC, rechargeable liquid ZABs were first assembled by using SAC-FeN-WPC as the air cathode catalyst (**Figure 4a**). The Pt/C+ $\text{RuO}_2$  mixture catalyst was tested for comparison. The open-circuit voltage of the SAC-FeN-WPC-based battery is 1.53 V, higher than that of Pt/C+ $\text{RuO}_2$ -based battery (1.33 V) (Figure S23). As shown in Figure 4b, the voltage drop of the ZAB with SAC-FeN-WPC at a constant current discharge and over a current density of 2-40  $\text{mA cm}^{-2}$  is very small, indicating that the ZAB has excellent discharge rate performance. When the current density is reduced to 2  $\text{mA cm}^{-2}$ , the discharge recovers reversibly, demonstrating the good rate performance and stability of SAC-FeN-WPC. The polarization curves in Figure 4c reveals a smaller charge-discharge voltage gap of SAC-FeN-WPC air electrode than that of Pt/C+ $\text{RuO}_2$ -based electrode at the same current density, suggesting a better ORR/OER activity of SAC-FeN-WPC air electrode. The corresponding peak power density of SAC-FeN-WPC-based battery is 152  $\text{mW cm}^{-2}$ , surpassing that of Pt/C+ $\text{IrO}_2$ -based battery (92.9  $\text{mW cm}^{-2}$ ). Furthermore, the SAC-FeN-WPC-based battery shows a specific capacity of 735.6  $\text{mAh g}^{-1}$  (normalized to the mass of zinc) at a current density of 10  $\text{mA cm}^{-2}$ . As compared to Pt/C+ $\text{RuO}_2$  reference ZAB (660.6  $\text{mAh g}^{-1}$ ), SAC-FeN-WPC based battery shows the

capacities of 763.8 and 707 mAh g<sup>-1</sup> at the current densities of 5 and 20 mA cm<sup>-2</sup>, respectively (Figure 4d). This indicates that SAC-FeN-WPC-based battery shows excellent rate performance. As shown in Figure 4e, the initial discharge and charge voltages of the SAC-FeN-WPC-based battery are 1.23 V and 2.04 V, respectively, contributing to an outstanding round-trip efficiency of 60.3%. For comparison, Pt/C+RuO<sub>2</sub>-based battery delivers an initial discharge and charge voltage of 1.21 V and 2.10 V with a low round-trip efficiency of 57.6%. After continuous cycling for 60 h, the gentle voltage change is observed for SAC-FeN-WPC air electrode. In comparison, the voltage gap of noble-metal Pt/C+RuO<sub>2</sub>-based battery increases gradually after 60h. Moreover, as shown in Figure 4f, two ZABs are connected in series to power a light-emitting diode (LED, ~3V). These results demonstrate the better cycling durability and bifunctional oxygen electrocatalytic reactivity of SAC-FeN-WPC, attributed to its robust porous structure and the existence of SACs.

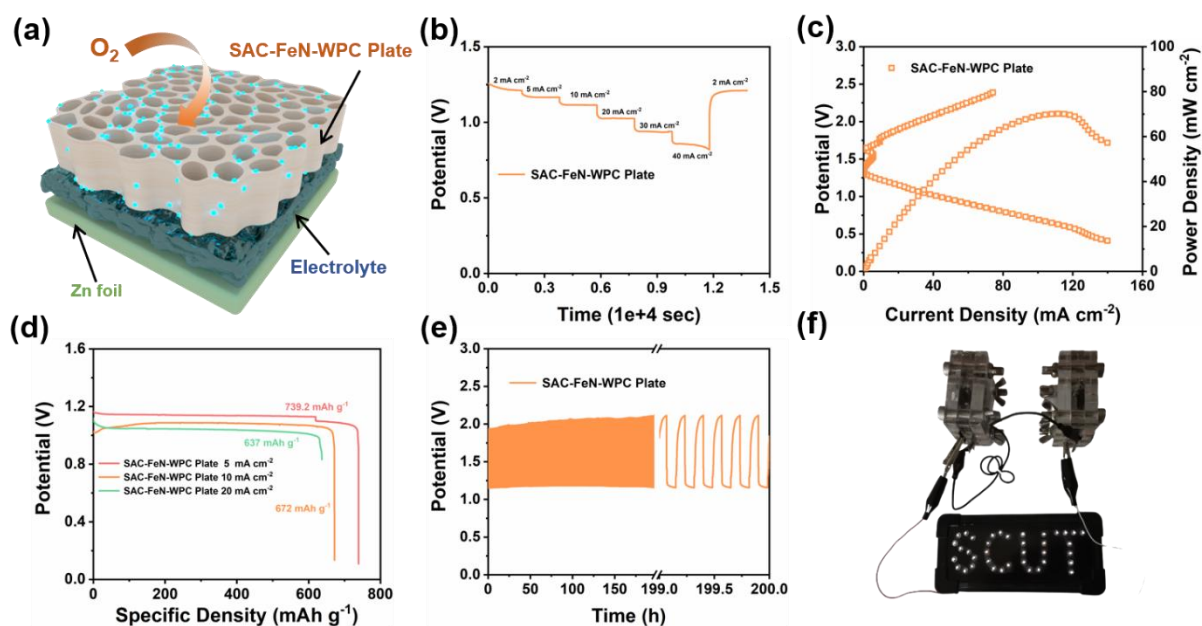


**Figure 4.** (a) Schematic of the aqueous rechargeable ZAB using SAC-FeN-WPC as the air cathode. (b) Discharge curves of the SAC-FeN-WPC-based ZAB at various current densities. (c) Charge and discharge polarization curves of the SAC-FeN-WPC and Pt/C+RuO<sub>2</sub> air electrodes for the rechargeable ZABs and the corresponding power density curves of ZABs.



**(d)** Galvanostatic discharge curves of the ZABs with the SAC-FeN-WPC powder and Pt/C + RuO<sub>2</sub> at different current densities. **(e)** Galvanostatic discharge/charge cycling curves at 5 mA cm<sup>-2</sup> of rechargeable ZABs with the SAC-FeN-WPC and Pt/C+RuO<sub>2</sub> cathodes. **(f)** Image of the LED powered by two liquid ZABs with SAC-FeN-WPC powder.

The quasi-solid-state ZABs were assembled using the SAC-FeN-WPC plate as the self-supporting electrode (**Figure 5a**). The open-circuit voltage of assembled battery is 1.32 V (Figure S24). As shown in Figure 5b, the gentle voltage drop indicates the satisfying discharge rate performance. Further, the discharge recovers reversibly when the current density back to 2 cm<sup>-2</sup>. The polarization curves of SAC-FeN-WPC plate-based quasi-solid-state ZAB (Figure 5c) display a narrow voltage gap, indicating the good charge-discharge ability of the ZAB. The corresponding peak power density of SAC-FeN-WPC plate-based quasi-solid-state ZABs is 70.2 mW cm<sup>-2</sup>. Figure 5d shows the discharge curve of SAC-FeN-WPC plate. SAC-FeN-WPC plate exhibits a specific capacity of 739.2, 672 and 637 mAh g<sup>-1</sup> (normalized to the mass of zinc) at the current density of 5, 10, and 20 mA cm<sup>-2</sup>, respectively. To further evaluate its durability, SAC-FeN-WPC plate-based ZAB was cycled for 200 h at 5 mA cm<sup>-2</sup>. As shown in Figure 5e, few voltage fluctuations are observed (0.77 V) with a round-trip efficiency of high up to 60%. Additionally, SAC-FeN-WPC plate-based ZAB can stably cycle for 118 h when the current density increases to 10 mA cm<sup>-2</sup> (Figure S25). As shown in Figure 5f, two SAC-FeN-WPC plate-based ZABs are connected in series to power an LED (~3 V). The results suggest the excellent activity and stability of the self-supported SAC-FeN-WPC plate electrode, which can be stemmed from its hierarchical porous structure and the presence of a large number of single-atom active sites in it and resulting in enhanced electron/mass diffusion.



**Figure 5.** SAC-FeN-WPC plate as the cathode for quasi-solid-state ZABs. **(a)** Schematic representation of a rechargeable quasi-solid-state ZAB with SAC-FeN-WPC plate cathode. **(b)** Discharge curves at various current densities. **(c)** Galvanostatic discharge/charge cycling curves and the corresponding power density of SAC-FeN-WPC plate-based quasi-solid-state battery. **(d)** Galvanostatic discharge curves of the ZABs with SAC-FeN-WPC plate at different current densities. **(e)** Galvanostatic discharge/charge cycling curves at  $5 \text{ mA cm}^{-2}$ . **(f)** Image of an LED powered by two rechargeable quasi-solid-state ZABs with SAC-FeN-WPC plate.

In summary, Lewis acid treatment is a convenient and effective method for simultaneously creating pores and in-situ growth of single-atom Fe on hierarchical and order pores of wood. Hydrolysis of Lewis acid effectively produces abundant microchannels in the cell walls, while the anchor of  $\text{Fe}^{3+}$  successfully introduces atomically dispersed FeN active species into the hierarchical structure. Due to these advantages, the as-prepared SAC-FeN-WPC catalyst shows excellent ORR and OER electrocatalytic performances and durability. More importantly, the plate SAC-FeN-WPC is capable of serving as a self-supported air electrode with outstanding rechargeability and high energy density. This work provides a new approach

to prepare plate single-atom catalysts as electrode materials on large scale for rechargeable ZABs via facile Lewis acid treatment.

## ASSOCIATED CONTENT

Supporting Information. Experimental details; Photos of batch processing of wood with  $\text{FeCl}_3$ ; SEM images of natural wood in the vertical direction; SEM images of  $\text{FeCl}_3$ -wood in the vertical direction; SEM images of N-WPC at various magnifications; SEM images of SAC-FeN-WPC at various magnifications; SEM images of  $\text{ZnCl}_2$ -Wood in the vertical direction; BET data; Raman spectra for WPC, N-WPC and SAC-FeN-WPC; N species and contents of SAC-FeN-WPC and N-WPC; SEM-EDS mapping of SAC-FeN-WPC; Aberration-corrected HAADF-STEM images of the SAC-FeN-WPC; High-resolution XPS spectrum of Fe 2p for SAC-FeN-WPC; Wavelet transform (WT) of the  $k^3$ -weighted Fe K-edge EXAFS; Fourier transform (FT) of the Fe K-edge EXAFS spectrum in  $R$  and  $k$  space and the corresponding fitting curve for SAC-FeN-WPC; Fourier transform (FT) of the Fe K-edge EXAFS spectrum in  $R$  and  $k$  space and the corresponding fitting curve for iron phthalocyanine (FePc); Fourier transform (FT) of the Fe K-edge EXAFS spectrum in  $R$  and  $k$  space and the corresponding fitting curve for Fe foil; Fourier transform (FT) of the Fe K-edge EXAFS spectrum in  $R$  and  $k$  space and the corresponding fitting curve for  $\text{Fe}_2\text{O}_3$ ; LSV ORR and OER curves for SAC-FeN-WPC before etching and SAC-FeN-WPC; LSV curves at different rotating rates for SAC-FeN-WPC and K-L plots at different potentials of SAC-FeN-WPC; LSV ORR curves at the rotating rate of 1600 rpm for SAC-FeN-WPC-2:1, SA-FeN-WPC-1:1, SA-FeN-WPC-1:2, SA-FeN-WPC-800, SA-FeN-WPC-900 and SA-FeN-WPC-1000. Chronoamperometric curves for the OER of SAC-FeN-WPC in an  $\text{O}_2$ -saturated 0.1 M KOH aqueous solution at 1600 rpm, N 1s high-resolution XPS spectra of SAC-FeN-WPC before and after OER i-t test, and the contents of different N species in SAC-FeN-WPC before and

after OER i-t test. LSV ORR curves at the rotating rate of 1600 rpm for SAC-FeN-WPC, AlCl<sub>3</sub>-N-WPC, and ZnCl<sub>2</sub>-N-WPC; LSV OER curves at the rotating rate of 1600 rpm for SA-FeN-WPC, AlCl<sub>3</sub>-N-WPC, and ZnCl<sub>2</sub>-N-WPC; Open circuit plots of SAC-FeN-WPC and Pt/C+RuO<sub>2</sub> based ZABs; Open-circuit plots of SAC-FeN-WPC plate; Galvanostatic discharge/charge cycling curves at 10 mA cm<sup>-2</sup>; BET surface area, pore volume and elemental contents of catalysis; Room temperature <sup>57</sup>Fe Mössbauer parameters for SAC-<sup>57</sup>FeN-WPC; Summary of the catalytic activities of the reported composite electrocatalysts in 0.1 M KOH;

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## CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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